



University of Al-Ameed
College of Medicine



Internal medicine department

Memory

الدكتور
علي لطيف السلطاني
شهادة البورد (الدكتوراه) في الطب النفسي
F.I.C.M.S(Psych.)

Learning Outcomes

At the end of this lecture, students will be able to:

- Articulate the concept of memory, including its role in encoding, storing, and retrieving information.
- Explain the three phases of memory: encoding, storage, and retrieval.
- Differentiate between various models of memory, including the Information Processing Approach and the Traditional Three-Stage Model.
- Distinguish between sensory memory, short-term (working) memory, and long-term memory, including their characteristics and functions.
- Identify the differences between declarative memory (semantic and episodic) and procedural (nondeclarative) memory.
- Summarize the brain regions and neurotransmitters involved in memory processes.
- Recognize how factors such as age, stress, sleep, medical conditions, and medication can impact memory.
- Utilize strategies to improve memory retention and recall, such as chunking, visualization, and spaced repetition.
- Identify the types of memory-related disorders, including amnesia and memory distortions, and their underlying causes.

Memory definition

- **Memory** is the information processing and cognitive ability to encode ,store and retrieved information.

Functions of Memory

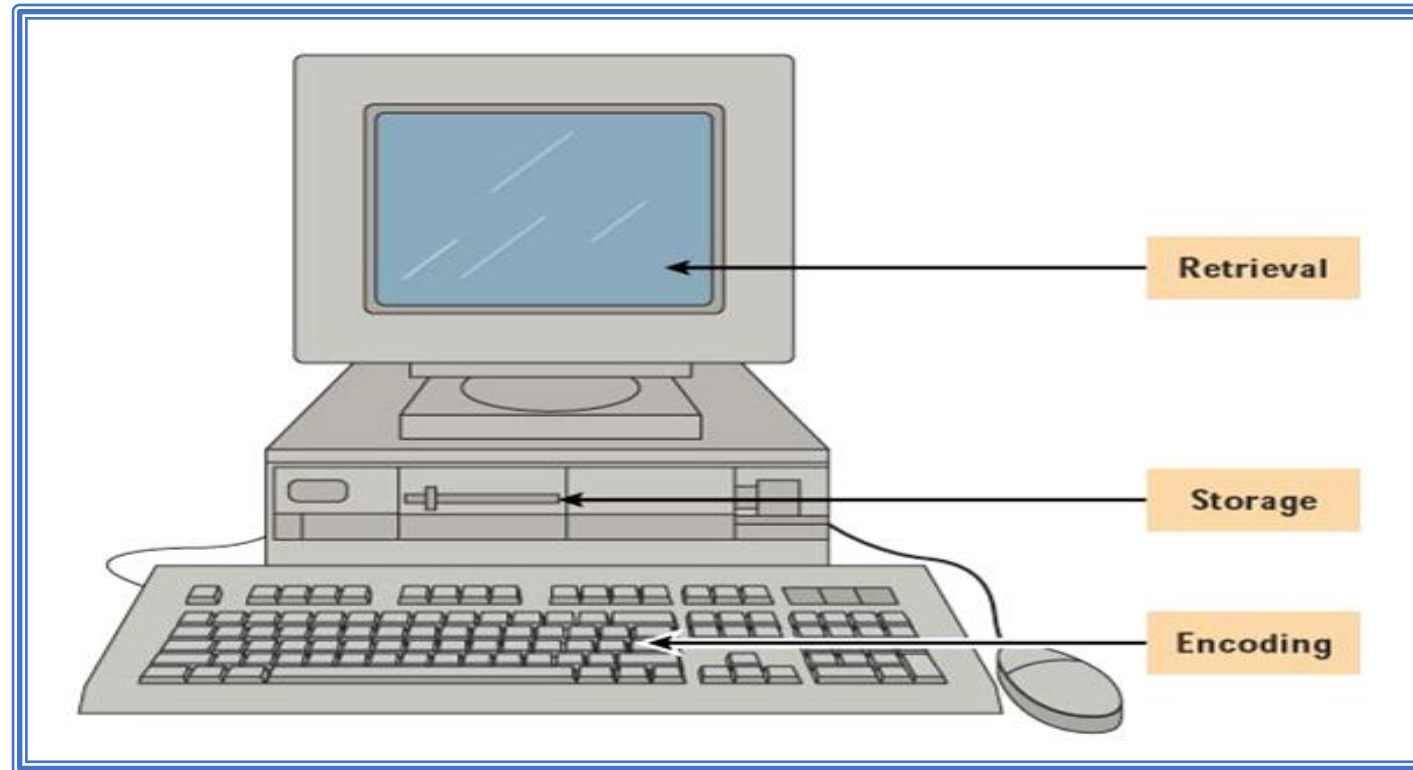
Three Phases of Memory

- **Encoding:** is active process of putting information into memory.
- When information comes into our memory system (from sensory input) it need to be changed into a form that the system can cope with ,so that it can be stored.
- **Storage:** is the process of holding encoding information in memory until time of retrieval.
- This concerns the nature of memory stores i.e. where the information is stored ,how long the memory last for(duration),how much can be stored at any time (capacity) and what kind of information is held.
- The way we stored information affect the way we retrieved it.
- It is either short term memory or long-term memory.

- **Retrieval** is the process of getting or recalling information that has been placed into short-term or long-term storage.(remembering).
- If we cannot remember something it may be because we are unable to retrieve it.

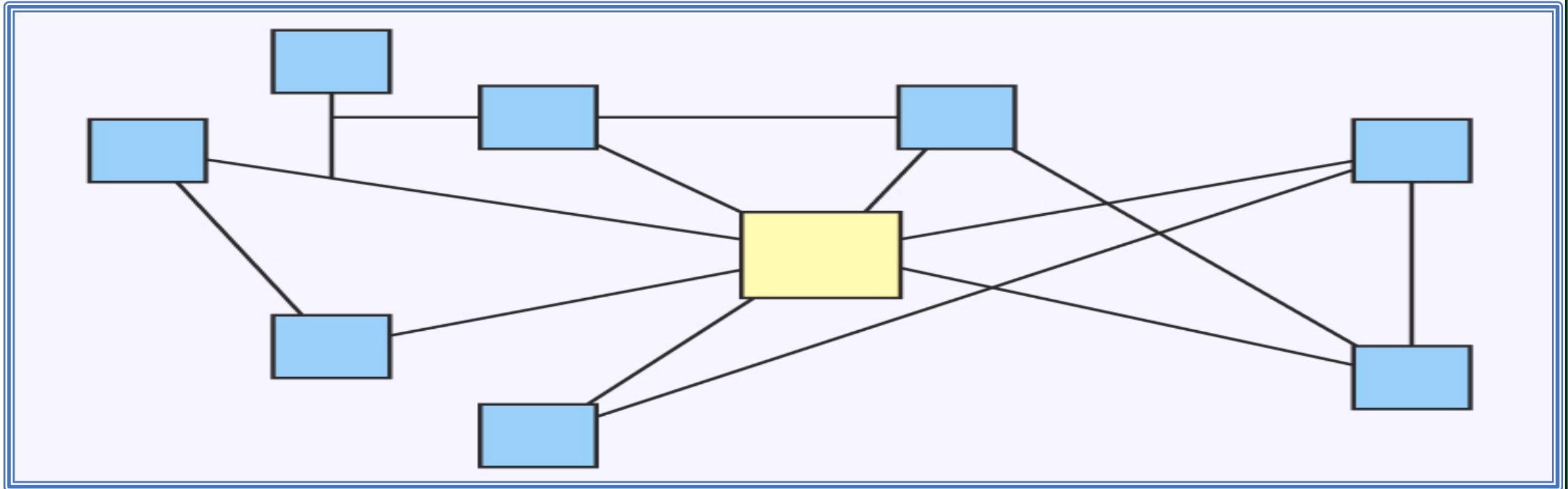
Four Memory Models

- **1. Information Processing Approach:** memory is a *process* analogous to a computer, which **encodes**, **stores**, and **retrieves** information



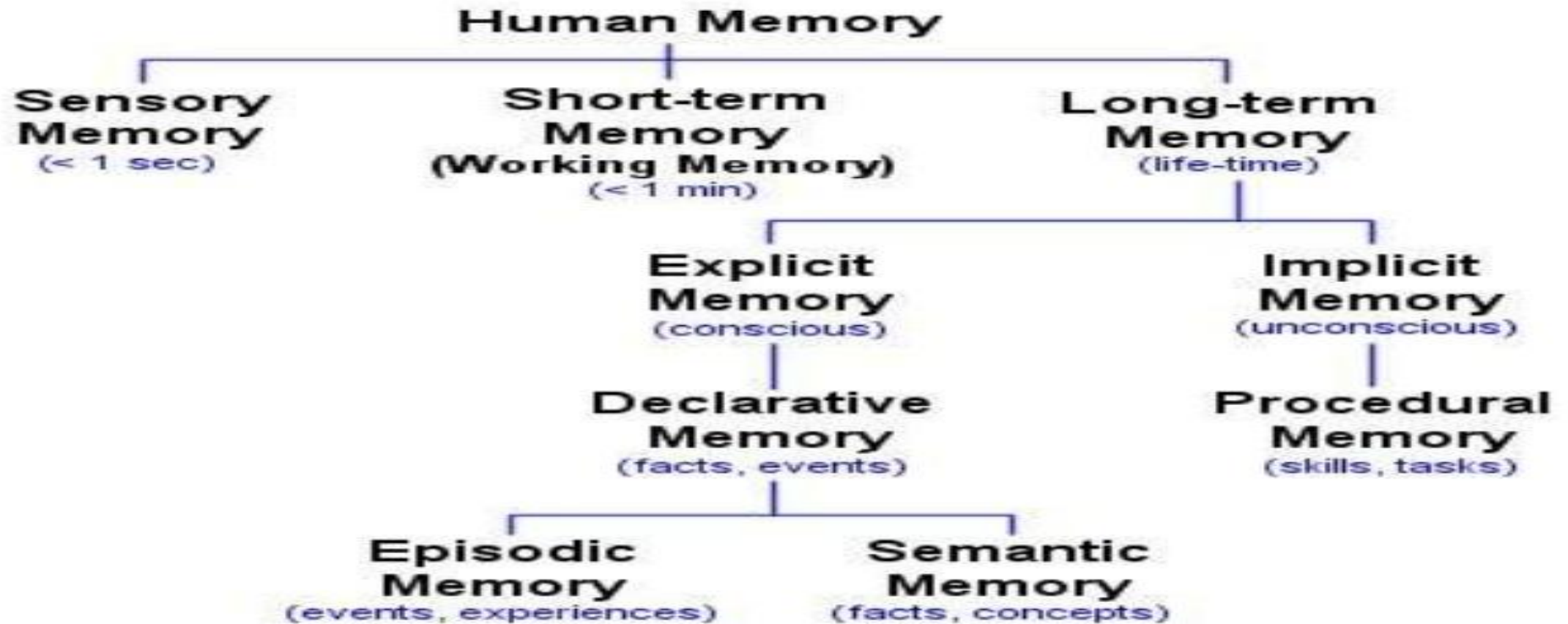
2. *Parallel Distributed Processing Model*:

memory is distributed across a network of interconnected units that work simultaneously (in a *parallel* fashion) to process information



3. *Levels of Processing Approach*: memory depends **on the degree or depth of mental processing occurring** when material is initially encountered
4. *Traditional Three-Stage Memory Model*: memory requires three different storage boxes to hold and process information for various lengths of time

Types of memory



- **1- Sensory memory** is the first system in the process of memory, the point at which information enters the nervous system through the sensory systems—eyes, ears, and so on.
- Think of it as a door that is open for a brief time.
- Looking through the door, one can see many people and objects, but only some of them will actually make it through the door itself.
- Sensory memory is a kind of door onto the world.
- The initial, momentary storage of information, *lasting only a second*.
- Information from the different sensory modalities is stored in separate sensory memories.
- *Iconic (visual)* memory is a visual sensory store with a short duration of less than one second.
- *Echoic (auditory)* memory is an auditory sensory store that lasts about 2 seconds.

Number of Letters Remembered

Delay in Seconds

0.0

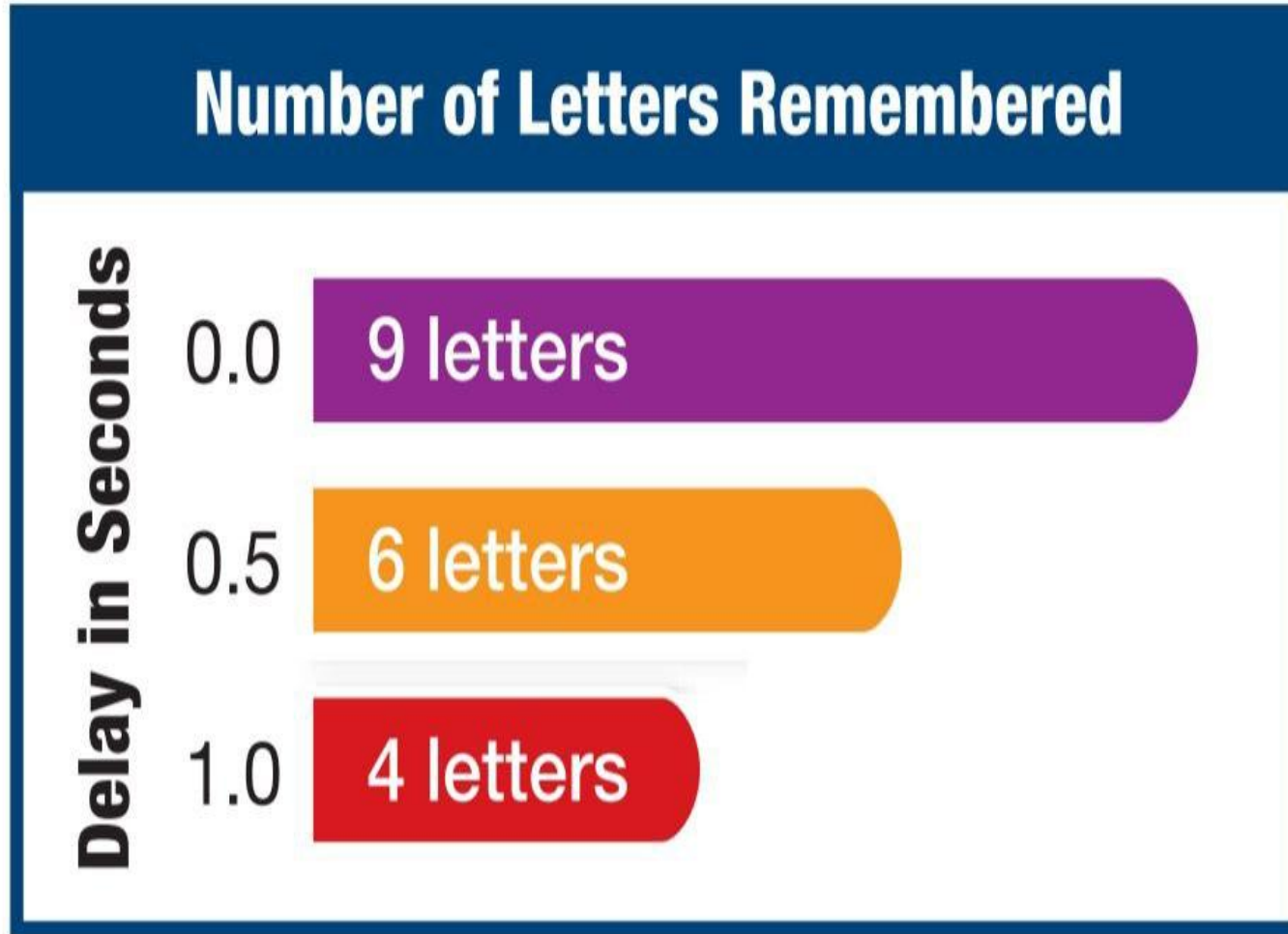
9 letters

0.5

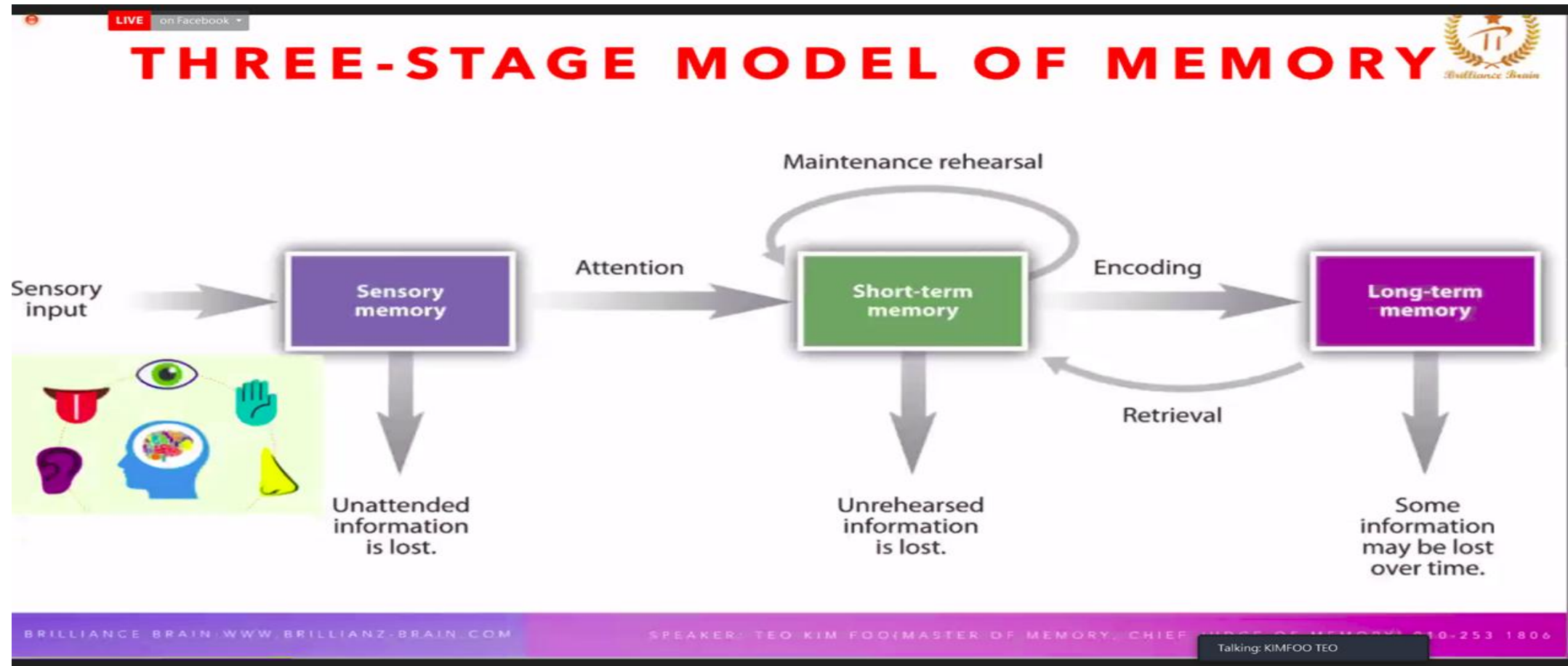
6 letters

1.0

4 letters



- If you **do not pay attention** to information in sensory memory, it is forgotten. If you **do pay attention** to particular information, such as the instructor's words, this information is automatically transferred into short-term memory.



2-Working memory

- Short-term memory, more recently called working memory, refers to a process that can hold a limited amount of information—an average of seven items—for a limited period of time—2 to 30 seconds.
- However, the relatively short duration can be lengthened by repeating or rehearsing the information.
- **Has a limited capacity** (5 to 9 bits of information.).
- Can only hold a small number of items.
- Capacity can be increased by grouping items into meaningful wholes called chunks. Limit is about 7 ± 2 chunks.

- There are three important points to remember about short-term memory: **1st—paying attention** transfers information into short term memory; **2nd—after a short time**, information disappears unless it is rehearsed; and **3rd—some information** is eventually transferred from short-term memory into permanent storage.
- **Maintenance rehearsal** refers to the practice of intentionally repeating or rehearsing information so that it remains longer in short-term memory.
- Coding can be acoustic, visual, or semantic.

TABLE 8.1 Digit-Span Test

Directions: Starting with the top sequence, read these numbers at a steady rate of one per second. Immediately after saying the last number in each series, signal the person to recall the numbers in order. Most people can recall a maximum sequence of 5 to 9 digits.

8 3 5 2

4 3 9 3 1

7 1 4 9 3 7

5 4 6 9 2 3 6

1 5 2 4 8 5 8 4

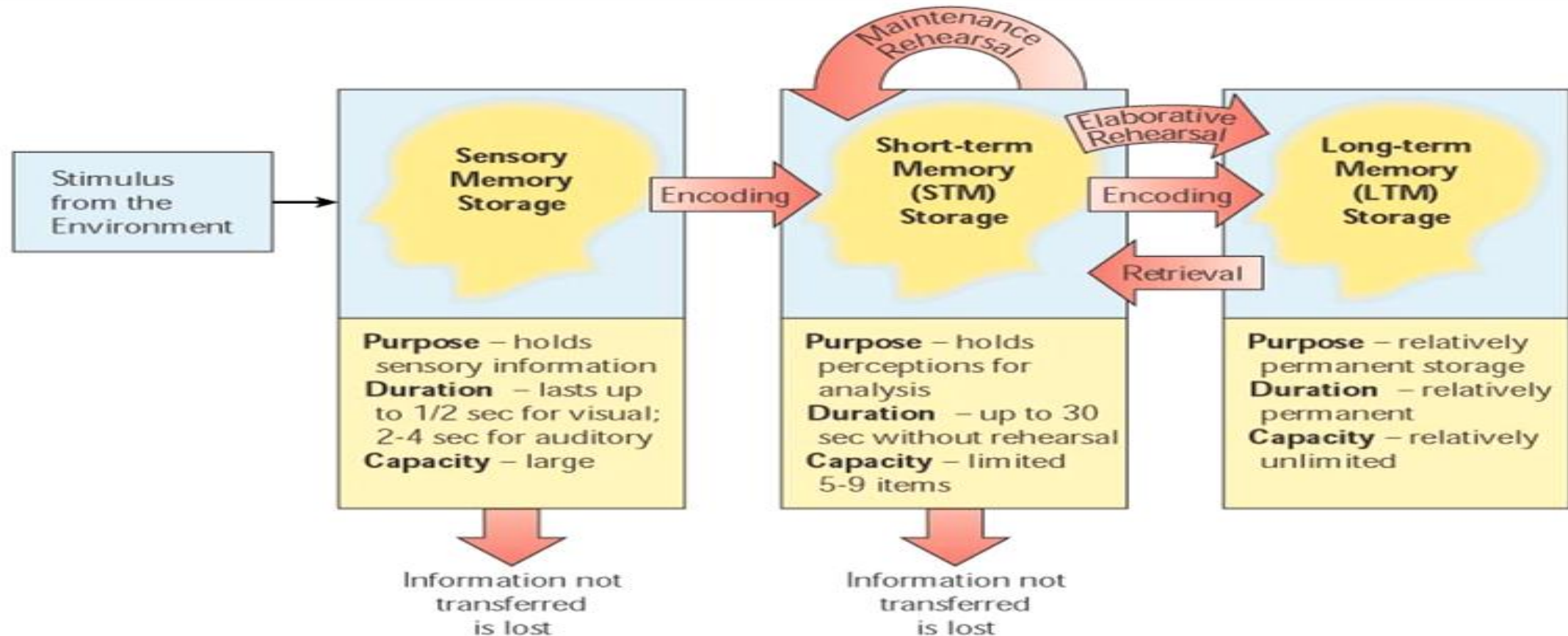
9 3 2 6 5 8 2 1 4

6 8 1 3 1 9 4 7 3 5

4 2 4 6 9 5 2 1 7 4 3

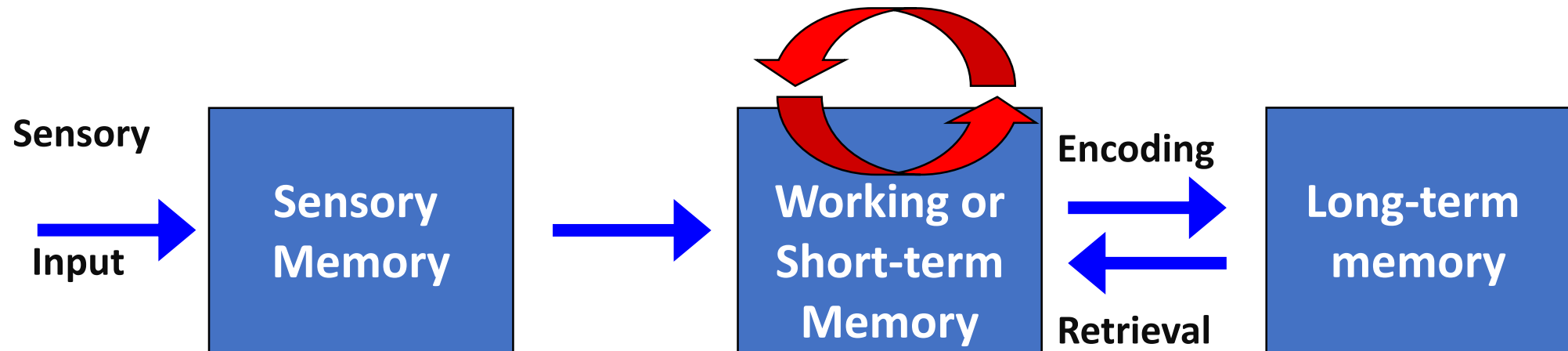
3 7 9 8 4 6 1 7 2 4 9 5

Diagram of Three-Stage Memory Model



3- Long-term memory- refers to the process of storing almost unlimited amounts of information over long periods of time with the potential of retrieving, or remembering, such information in the future.

- **Unlimited capacity and duration.**
- Once information passes from sensory to working memory, it can be encoded into long-term memory



- **Encoding**—You place information in long-term memory through a process called encoding.
- **Encoding** is the process of transferring information from short-term to long-term memory by paying attention to it, repeating or rehearsing it, or forming new associations.
- **Retrieval**—process that controls flow of information from long-term to working memory store.
- The holdings of your long-term memory, like those of a library, must be organized if they are to be available when you want to retrieve them. The more effectively we encode material into long term memory, the greater the likelihood of retrieving it.

Varieties of Long-Term Memory

Explicit (Declarative) Memory

Memory with conscious recall

Semantic Memory

Facts and general knowledge (e.g., bananas are yellow, 12 months in a year, spiders have eight legs)



Episodic Memory

Personal experiences and events (e.g., your high school graduation, the birth of your first child)



Implicit (Nondeclarative) Memory

Memory without conscious recall

Procedural Memory

Motor skills and habits (e.g., how to drive a car, brush your teeth, ride a bike)



Classically Conditioned Memory

Conditioned responses to conditioned stimuli (e.g., phobias, some aspects of prejudice, and other attitudes)



Priming

Earlier exposure facilitates retrieval (e.g., heightened fears after reading a scary novel)



1- Declarative memory involves memories for facts or events, such as scenes, stories, words, conversations, faces, or daily events. We are aware of and can recall, or retrieve, these kinds of memories.

- There are two kinds of declarative memory—semantic and episodic
- **Semantic memory** is a type of declarative memory and involves knowledge of facts, concepts, words, definitions, and language rules.
- Most of what you **learn** in classes (facts, terms, definitions) goes into semantic memory.
- **Episodic memory** is a type of declarative memory and involves knowledge of specific events,
 - personal experiences (episodes), or activities, such as naming or describing favorite restaurants, movies, songs, habits, or hobbies.
 - Most of your college activities and experiences go into episodic memory

2- Procedural or Nondeclarative Memory are stored in procedural memory. Procedural memory, also called nondeclarative memory, involves memories for motor skills (playing tennis), some cognitive skills (learning to read), and emotional behaviors learned through classical conditioning (fear of spiders).

We cannot recall or retrieve procedural memories.

Anatomical areas involved in memory

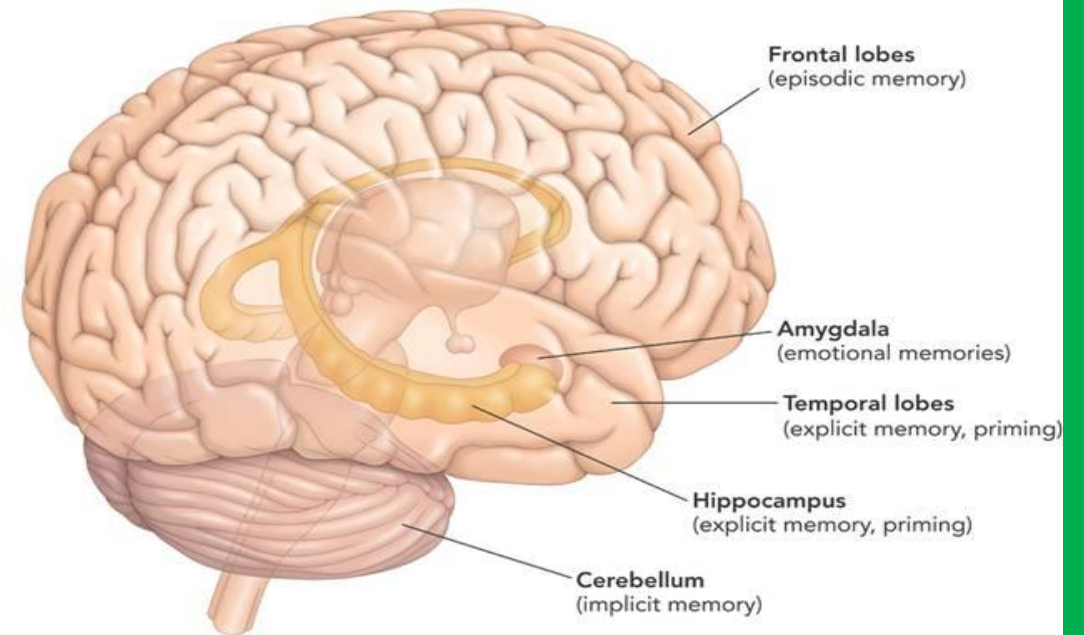
Hippocampus

Amygdala

Prefrontal Cortex

Basal Ganglia

Cerebellum



Neurotransmitters involved in memory

- Acetyl choline
- serotonin
- Dopamine
- Nor-Epinephrine
- Some neuropeptides
- others (protein, DNA, RNA)

Factors contribute to forgetfulness

- **1. Age:** As people get older, they may experience age-related memory decline, often referred to as age-associated memory impairment.
 - It is a normal part of the aging process and typically involves mild forgetfulness, such as forgetting names or misplacing objects.
- **2. Stress and anxiety:** High levels of stress and anxiety can interfere with concentration and memory retrieval, making it more difficult to remember things.
- **Stress hormones** can affect the hippocampus, a brain region involved in memory formation and retrieval.
- **3. Lack of sleep:** Sleep deprivation can impair cognitive function and memory.
 - When you don't get enough sleep, it becomes challenging to concentrate and retain information.
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- **4. Medical conditions:** Certain medical conditions can cause forgetfulness. Examples include **thyroid disorders, vitamin deficiencies** (such as vitamin B12), **infections, head injuries**, and neurological disorders like **Alzheimer's disease** or dementia.
- **5. Medications and substance abuse:** Some medications, such as those used to treat anxiety, depression, or sleep disorders, can have side effects that affect memory and cognition.
- Substance abuse, including **alcohol** and drug use, can also impair memory function

Memory enhancement strategies

- **1. Pay attention:** Give your full attention to the information you want to remember.
- **2. Use mnemonic devices:** Mnemonic devices are memory aids that can help you remember information.
- **3. Chunking:** Break down large amounts of information into smaller, manageable chunks.
- **4. Visualization:** Create mental images or visual representations of the information you want to remember.
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- **5. Association:** Connect new information to something you already know.
- **6. Practice retrieval:** Actively practice retrieving information from your memory.
- **7. Spaced repetition:** Distribute your study or practice sessions over time rather than cramming all at once.
- **8. Use multi-sensory techniques:** Engage multiple senses while learning.
- **9. Sleep well:** Sufficient sleep is crucial for memory consolidation.
- **10. Stay organized:** Keep your study materials, notes, and information organized.

11. Teach someone else: Explaining concepts or teaching someone else

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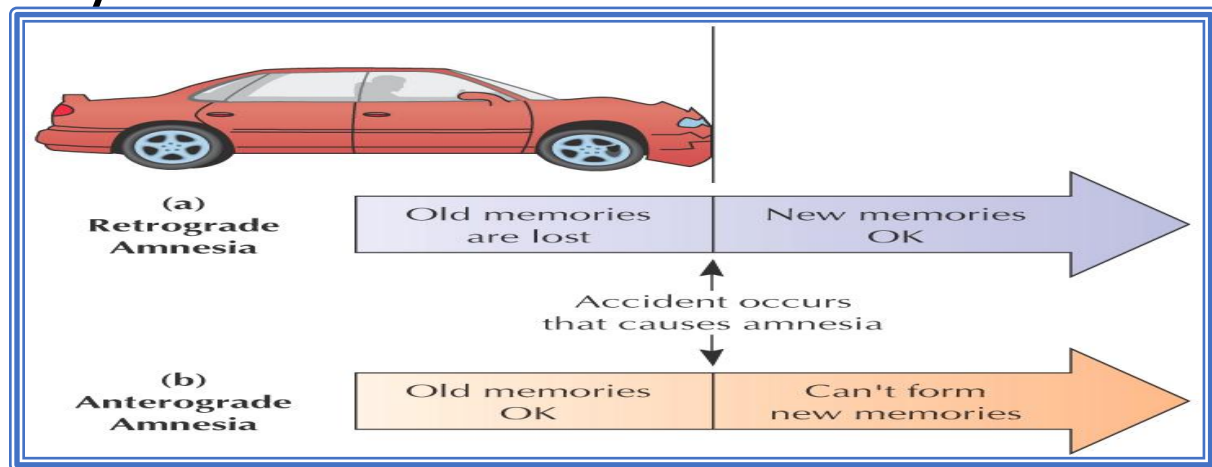
12. Stay mentally active: Regularly engage in activities that challenge your brain.

13. Manage stress: High levels of stress can impair memory and cognitive function.

Psychopathology of memory

1. **Amnesia**: Amnesia, involves the partial or complete loss of memory. It can be caused by physical injuries, medical conditions, psychological trauma, or certain psychiatric disorders.

- **Retrograde amnesia**: old memories lost
- **Anterograde amnesia**: new memories lost
- **Transient Global Amnesia**: Transient global amnesia is a temporary form of amnesia that usually lasts for a few hours.



- **2. Memory Intrusions:** Memory intrusions refer to the **involuntary and unwanted recall of distressing or traumatic memories**. They are commonly associated with conditions like **PTSD**.
- **3. Memory Distortions:** Memory distortions involve **inaccuracies or alterations in memory recall**. Memory distortions are commonly observed in conditions like **depression, anxiety** disorders, and schizophrenia.

Communication Skills

- This lecture supports the Communication Skills by helping students understand how memory processes influence patient understanding, recall, and adherence to medical advice. Effective communication in clinical settings depends on how information is encoded, stored, and retrieved by patients during consultations.
- The lecture highlights the stages of memory—encoding, storage, and retrieval—and shows that attention, repetition, and meaningful organization improve retention, while stress, sleep problems, and illness can impair recall .
- These concepts guide clinicians in delivering information clearly and reinforcing key messages.
- Applying this knowledge strengthens communication by enabling students to:
 - present information in clear, structured, and memorable ways
 - use repetition, chunking, and visual explanations to improve patient understanding
 - check patient recall and comprehension during consultations
 - adapt communication for patients with memory difficulties, stress, or illness